Total No. of Questions : 8]

SEAT No. :

PF-5497

[Total No. of Pages : 2

[6627]-374

B.E. (AI & DS)

DATA MODELING & VISUALIZATION

(2019 Pattern) (Semester - VII) (417522)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Draw figures wherever necessary.***
4) ***Figures to right indicate full marks.***
5) ***Assume suitable data if necessary.***

Q1) a) Define static graphics. Discuss the features of complete plot. **[6]**

b) Explain the concept of scientific design choices in visualization. **[6]**

c) Differentiate between presentation Graphics and Exploratory Graphics with examples. **[5]**

OR

Q2) a) Discuss how data handling techniques affect visualization accuracy and performance. **[6]**

b) Write short notes on output formats and speed issues. **[6]**

c) Explain 3-D plots in modeling data Visualization. **[5]**

Q3) a) Describe the pulling under constraints model in graph visualization. **[6]**

b) Discuss various function available in matplotlib. **[6]**

c) What is multidimensional scaling (MDS) and how MDS helps in visulizing high dimensional data. **[6]**

OR

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Q4) a) Write short notes on Reshaping and pivoting. **[6]**

b) Describe any two layout techniques used in graph visualization with examples. **[6]**

c) What is seaborn? Discuss the advantages over matplotlib. **[6]**

Q5) a) Explain the concept of Resampling and its role in data analysis and smoothing. **[8]**

b) Write a short note on: **[10]**

i) Pivot tables

ii) Periods and periods arithmetic

OR

Q6) a) Explain Cross Tabulation and its uses with suitable example? **[9]**

b) Explain Time Series Basics with a suitable example and how time indexed data differs from regular datasets. **[9]**

Q7) a) What are PET images? Explain the process of PET image formation. **[9]**

b) Explain the various Data Reconstruction Techniques used to assist in data modeling and visualization. **[8]**

OR

Q8) a) Explain Lucid chart and its use in data visualization and database modeling. mention advantages of Lucidchart over traditional ER diagram tools. **[9]**

b) Write a short notes on: **[8]**

i) ER / Studio

ii) Analysis of medical images.

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